Machine Learning

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Professor

- Simão Paredes [sparedes@isec.pt] Theoretical / Practical

Objectives

- Explore the fundamental concepts in the area of Machine Learning, namely:
  - Understand the main phases of a real project based on Machine Learning
  - Analysis and preparation of a dataset
  - Classification (metrics, validation strategies, ...)
  - Most relevant algorithms in supervised learning
  - Most relevant algorithms in unsupervised learning

Theoretical Component

- Chap.1 – Introduction
  - Data Mining / Machine Learning
  - Types of Learning
  - Classification
  - Regression
  - ML Main Challenges
  - Notation
- Chap.2 – Data: Fundamental Concepts
  - Types of Data
  - Descriptive Statistics
  - Random Variable
  - Data Distributions
  - Data Cleaning

Theoretical Component

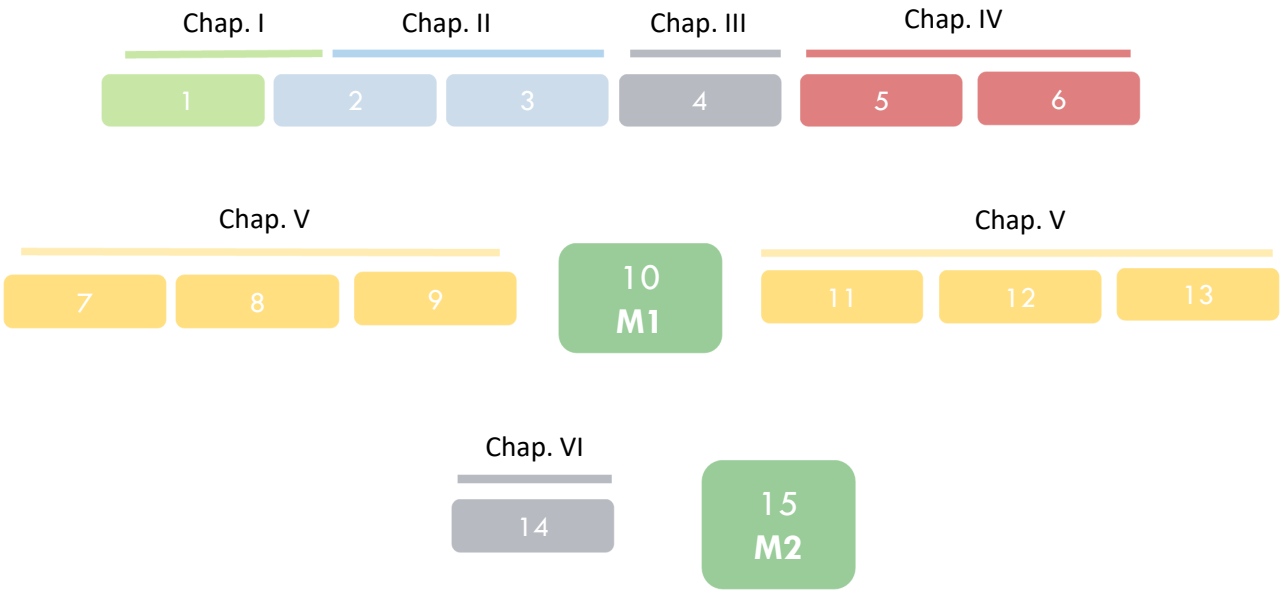
- Chap. 3 – Main Stages of a Project
  - Problem identification/contextualization
  - Data acquisition
  - Data visualization/preparation
  - Model selection and training
  - Model tuning/adjustment
  - Monitoring and Maintenance
- Chap. 4 – Classification
  - Evaluation Metrics
  - Validation Strategies
  - Error Analysis

Theoretical Component

- Chap. 5 – Supervised Learning
  - Linear Regression
  - Logistic Regression
  - Naïve Bayes
  - Decision Trees
  - Support Vector Machine
  - KNN-nearest Neighbor
  - Random Forest
  - XGBoost
  - Neural Networks
- Chap. 6 – Unsupervised Learning
  - Clustering
    - K-means
    - Subtractive Clustering
  - Principal Component Analysis

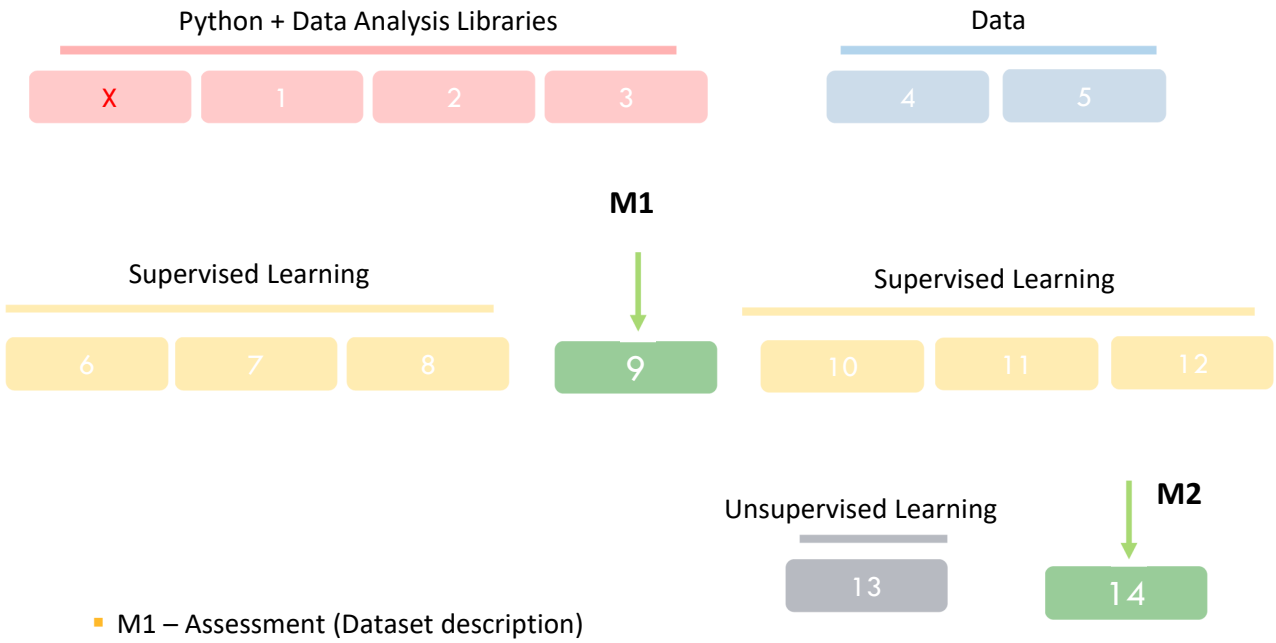
Theoretical Component

- Theoretical class (2hours/week)



Practical Component

- Practical classes (2 hours/week)



- M1 – Assessment (Dataset description)
- M2 – Assessment – Classifiers Comparison SL + approaches UL

Assessment

- The final grade of the course is obtained through the following expression:

Final grade = 0.6 * Exam + 0.4 * Practical Assessment

- Components have different weights:
  - Exam: 12 values
  - Practical Work: 8 values
- Exam
  - A written exam, which will cover the content taught in theoretical and practical classes.
  - It is mandatory to obtain a grade equal to or greater than 35%.
    - Failure to achieve this minimum grade will result in failing the course.
  - The exam will be held during the exams' periods.

Assessment

- Practical Assessment
  - To be implemented based on Python.
  - Students must form groups of 2 students.
    - Two assessment moments:
      - M1 – 3 values
      - M2 – 5 values
  - In each assessment students must:
    - Present the respective work
      - Presentation (max. 10 minutes)
    - Prepare a small document that presents the work in an objective and condensed way (max. 10 pages)
    - The developed work must be submitted before the presentation in class.
      - All submission/presentation dates will be announced/confirmed in due time.

Assessment

- Practical Assessment
  - The grade for the practical work will be considered in all exam periods held during the academic year.
  - There will be no new opportunity to complete/submit the assessment elements during the academic year.

Bibliography

- Theoretical classes presentations
  - inforestudante
  - a********@isec.pt (Very Important!!)
- References
  - [1] Geron A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems 2rd Edition. ISBN: 978-1-098125-974, O'Reilly (ISEC: 1A-4-219)
  - [2] Chen, D. (2018) Pandas for Everyone, Python Data Analysis, ISBN:13-978-0-13-454693-3, Addison-Wesley, (ISEC: 1A -19-26)
  - [3] Rashka S, Mirjalili V. (2019). Python Machine Learning, 3rd Edition. ISBN: 978-1-78995-575-0, Packt. (ISEC: 1A – 4-187)
  - [4] Tarek A. Atwan (2022) Time Series Analysis with Python, ISBN: 978-1-801107-554-1, Packt Publishing (ISEC: 1A-4-212)
  - [5] Watt J., Borhani R., Katsaggelos A. (2020). Machine Learning Refined, 2nd Ed. ISBN: 978-1-108-48072-7, Cambridge University Pack. (ISEC: 1A – 4- 200)

Bibliography

- Other Resources:
  - https://www.kaggle.com/
  - https://towardsdatascience.com
  - https://realpython.com/
  - https://scipy-lectures.org/
  - ….